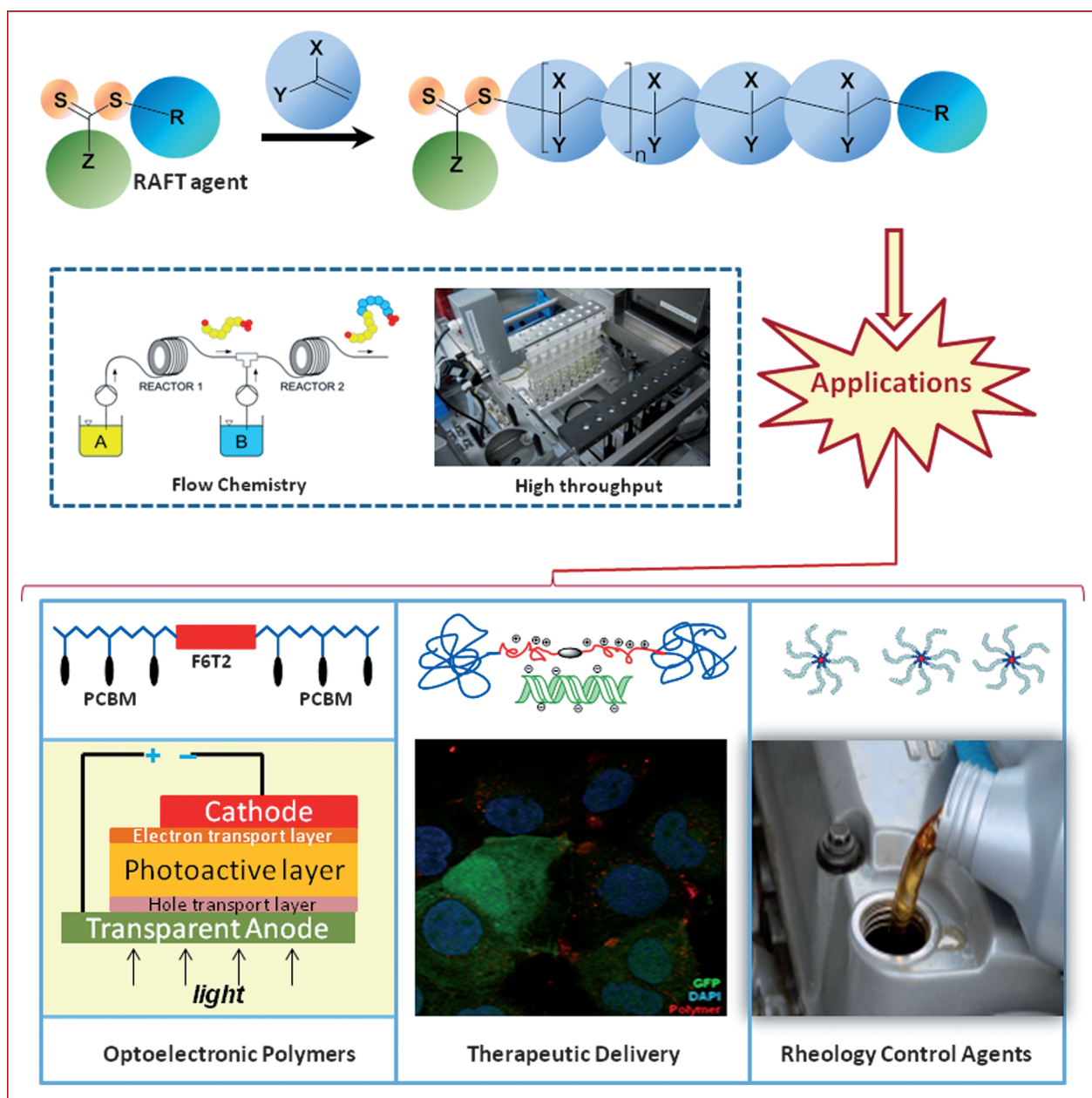


RAFT Polymerization and Some of its Applications

Graeme Moad,* Ezio Rizzardo,* and San H. Thang*[a]



Abstract: Reversible addition–fragmentation chain transfer (RAFT) is one of the most robust and versatile methods for controlling radical polymerization. With appropriate selection of the RAFT agent for the monomers and reaction conditions, it is applicable to the majority of monomers subject to radical polymerization. The process can be used in the synthesis of well-defined homo-, gradient, diblock, triblock, and star polymers and more complex architectures, which include microgels and polymer brushes. In this Focus Review we describe how the devel-

opment of RAFT and RAFT application has been facilitated by the adoption of continuous flow techniques using tubular reactors and through the use of high-throughput methodology. Applications described include the use of RAFT in the preparation of polymers for optoelectronics, block copolymer therapeutics, and star polymer rheology control agents.

Keywords: mechanism • polymerization • radicals • RAFT • synthesis design

Introduction

Radical polymerization is one of the most widely used processes for the commercial production of high-molar-weight polymers. The main factors responsible for the technique's preeminent position are its broad applicability to monomers and reaction conditions. While radical polymerization is simple to implement and inexpensive in relation to competitive technologies, the conventional process has severe limitations with respect to the degree of control that can be asserted over molar weight distribution, copolymer composition, and macromolecular architecture.^[1]

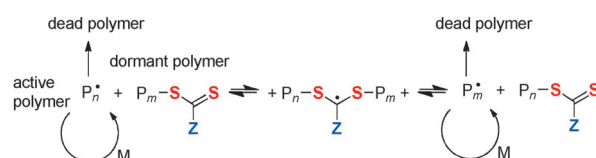
This situation has been redressed with the advent of techniques for reversible deactivation radical polymerization (RDRP),^[2] which provide radical polymerization with most of the attributes of living polymerization while retaining many of the benefits of the conventional process. These attributes include that all chains grow simultaneously, albeit intermittently, that block copolymers can be prepared by addition of further monomer, and that end-group functionality is preserved. The basic requirement of RDRP, and the feature that distinguishes RDRP from classical living polymerization, is that the majority of living chains are maintained in a dormant form. Living behavior is then a consequence of the facile equilibrium between dormant and active propagating chains. The concentration and cumulative lifetime of active chains under the polymerization conditions is typically similar to or lower than that observed in conventional radical polymerization.^[3]

Papers on the development and use of RDRP methods now account for more than half of the literature on radical polymerization and its applications, and all of the growth in this field over the last two decades.^[1] More than a third of these RDRP papers now relate to RAFT (reversible addition–fragmentation chain transfer) polymerization.^[4]

1. RAFT Polymerization

1.1. RAFT Mechanisms

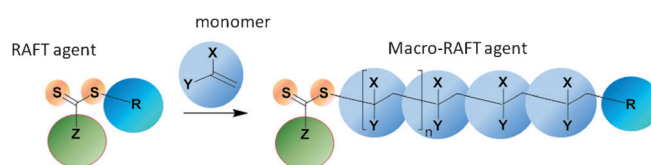
RAFT polymerization was developed at CSIRO in the mid-1990s and has proved to be a robust and versatile process that is applicable to the majority of monomers subject to radical polymerization.^[4,5] The activation/deactivation process in RAFT polymerization involves degenerate chain transfer (Scheme 1).^[6] The chain-transfer step is termed de-



Scheme 1. Mechanism for RDRP with reversible activation/deactivation by degenerate chain transfer.

generate because the process involves only an exchange of functionality and the only distinction between the species on the two sides of the equilibrium is their degree of polymerization (n and m) and, in an effective process, these will be similar.

The overall RAFT process, in which an initial RAFT agent (**1**) is converted into a polymeric macro-RAFT agent (**2**) by sequential insertion of monomer units into a C–S bond, is shown in Scheme 2.



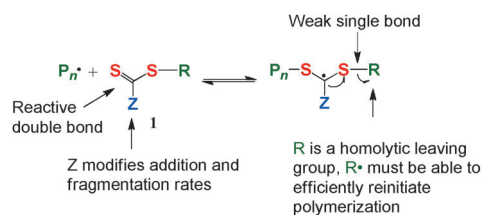
Scheme 2. Overall RAFT process.

A successful outcome in RAFT polymerization depends on the selection of the RAFT agent (**1**) for the specific monomers and reaction conditions. Guidelines for selection of RAFT agents have been proposed such that the polymerization of most monomers can be well controlled to provide

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minimal retardation and a high fraction of living chains.^[5a-c,7] The required RAFT agent properties are shown in Scheme 3.

A wide range of thiocarbonylthio compounds (**1**) has now been reported for use in the process.^[8] The effectiveness of a RAFT agent depends on the monomer being polymerized and is determined by the properties of the radical leaving group 'R' and the activating group Z. Some examples of RAFT agents are **3–17**.



Scheme 3. Properties of effective RAFT agents.



Graeme Moad obtained his BSc (Hons, First Class) and PhD from Adelaide University in organic free radical chemistry. He joined CSIRO in 1979 where he is currently a Chief Research Scientist. Dr. Moad is (co)author of over 150 publications, co-inventor of 34 patent families, and co-author of the book "The Chemistry of Radical Polymerization". His research interests lie in the fields of polymerization mechanism, and polymer design and synthesis. In recognition of his work Dr. Moad was recently awarded the RACI's Battaerd-Jordan Polymer Medal. Dr. Moad is a titular member of the IUPAC Polymer Division and a Fellow of the RACI and the Australian Academy of Science.



Ezio Rizzardo (FRACI, FTSE, FAA, FRS) received his PhD from the University of Sydney in 1969 and joined CSIRO in 1976 after postdoctoral research at Rice University, RIMAC, and the Australian National University. His CSIRO research has focused on developing methods for controlling free radical polymerization. He is co-author of some 200 journal papers with over 14000 citations, and co-inventor on 44 worldwide patents. He has received a number of awards including the RACI Australian Polymer Medal, the CSIRO Chairman's Gold Medal, and an Australian Government Centenary Medal. In 2011 he was the co-recipient of the Australian Prime Minister's Prize for Science.



San Thang completed his PhD in chemistry at Griffith University in 1987. Currently, he is a Chief Research Scientist at CSIRO Materials Science and Engineering; and an Adjunct Professor at Monash University. His research focuses on the interface between biology and polymer chemistry. Dr. Thang has published over 100 papers which have to-date received over 10000 citations. He is responsible for several key inventions in the area of controlled/living radical polymerization; significantly, he is a co-inventor of the RAFT process. He is a Fellow of the Australian Academy of Technological Science and Engineering, and a Fellow of the Royal Australian Chemical Institute.

The 'Z' group is chosen to convey appropriate reactivity to the thiocarbonyl double bond of the RAFT agent (**1**) and the intermediate species formed by radical addition.^[9] RAFT agents such as dithioesters (**1**, Z = aryl or alkyl, e.g., **3–6**) or trithiocarbonates (**1**, Z = alkylthio, e.g., **7–12**) are suitable for controlling the polymerization of 'more activated' monomers (MAMs; e.g., methyl methacrylate (MMA), styrene (St), methyl acrylate (MA), acrylamide (AM), and acrylonitrile (AN)) but inhibit or retard polymerizations of 'less activated' monomers (LAMs, e.g., vinyl acetate (VAc), N-vinylpyrrolidone (NVP), and N-vinylcarbazole (NVC)). RAFT agents such as xanthates (**1**, Z = alkoxy, e.g., **13, 14**) and N,N-dialkyl- or N-alkyl-N-aryl dithiocarbamates (**1**, Z = N,N-dialkylamino or N-alkyl-N-arylamino, e.g., **15, 16**) can be suitable for controlling polymerizations of LAMs but are ineffective with MAMs. The dithiocarbamate **16** is switchable such that, in neutral form, it controls the polymerizations of LAMs but, in its protonated form (**17**), provides effective control over the polymerization of MAMs.^[10]

The effect of an added RAFT agent on molecular weight distribution is illustrated in Figure 1^[6] and Figure 2.^[11] For the RAFT co-polymerization of styrene (St)–acrylonitrile (AN) with cumyl dithiobenzoate (**3**) and thermal initiation at 100°C, the number average molar masses (M_n) are observed to increase in a systematic way with time and mono-

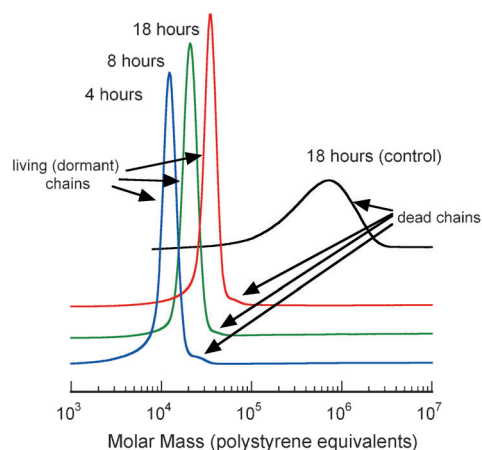


Figure 1. Molar mass distributions for poly(St-co-AN) polymerized by heating St and AN (62:38 molar ratio) at 100°C in the presence of cumyl dithiobenzoate (**3**) (0.0123 M) after 4 h (M_n 20100; $\bar{D}$ 1.04), 8 h (M_n 33000; $\bar{D}$ 1.05), and 18 h (M_n 51400; $\bar{D}$ 1.07). The molecular weight distribution for a control polymerization performed without cumyl dithiobenzoate is also shown (M_n 424000, $\bar{D}$ 1.70).^[6]

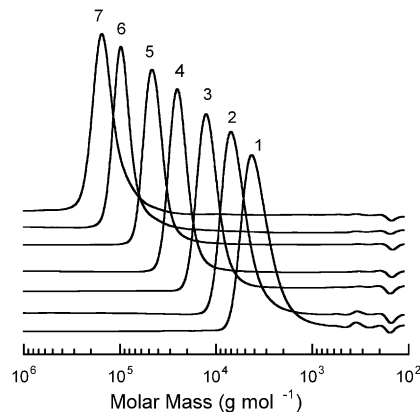
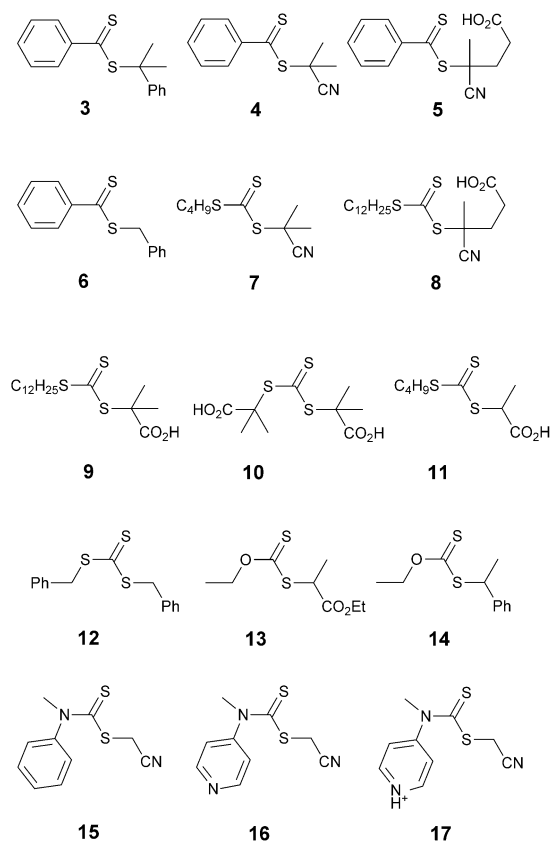


Figure 2. Gel permeation chromatography (GPC) traces of poly(MMA) formed by RAFT polymerization of MMA (6.55 M in benzene) with various concentrations of RAFT agent **8** and with azobis(1-cyclohexanenitrile) (0.0018 M) as initiator and a reaction time of 6 h at 90 °C. For trace 1, $[8]_0$ 19.92×10^2 M, M_n 2870, $\bar{D}$ 1.18, $M_n(\text{calc})$ 3000, 80 % conversion; trace 2, $[8]_0$ 9.96×10^2 M, M_n 5040, $\bar{D}$ 1.14, $M_n(\text{calc})$ 5600, 80 % conversion; trace 3, $[8]_0$ 4.95×10^2 M, M_n 9940, $\bar{D}$ 1.12, $M_n(\text{calc})$ 10400, 79 % conversion; trace 4, $[8]_0$ 2.48×10^2 M, M_n 21800, $\bar{D}$ 1.13, $M_n(\text{calc})$ 22600, 91 % conversion; trace 5, $[8]_0$ 1.24×10^2 M, M_n 41100, $\bar{D}$ 1.14, $M_n(\text{calc})$ 45300, > 99 % conversion; trace 6, $[8]_0$ 0.61×10^2 M, M_n 80900, $\bar{D}$ 1.13, $M_n(\text{calc})$ 80100, > 99 % conversion; trace 7, $[8]_0$ 0.32×10^2 M, M_n 126000, $\bar{D}$ 1.15, $M_n(\text{calc})$ 125000, > 99 % conversion. Figure reproduced from ref. [11] with permission.

mer conversion. The breadths of the molar mass distributions are narrow, which is reflected in the dispersities being very low ($\bar{D} < 1.1$). A small number of dead chains formed by termination is manifested as a molar mass shoulder and a low-molar-mass tail in the molar mass distributions (Figure 1). The amount of termination is consistent with that predicted based on the rate of initiation, the reaction time, and the monomer conversion.

For RAFT polymerization of methyl methacrylate (MMA) (Figure 2), a series of high conversions (80–100 %) were carried out at 90 °C with trithiocarbonate RAFT agent **8** and ACHN (1,1'-azobis(1-cyclohexanenitrile)) initiator, using an about 60-fold range of initial RAFT agent concentrations.^[11] The values of M_n , which range from 2870 to 126000, vary in accord with the concentrations of RAFT agent used. All samples have narrow molecular weight distributions ($\bar{D} < 1.2$). Termination, while undeniably present, cannot be detected.

The living character necessary for the synthesis of highly pure block copolymers, star polymers, and more complex architectures, such as those shown in Figure 3, is dependent on the presence of the RAFT end-group.^[3–5] End-group fidelity can be maximized by minimizing initiator concentration and choosing reaction conditions such that the target molar mass is substantially lower than that of polymer formed in the absence of RAFT agent.^[12] Additionally, polymerization should ideally be stopped before complete monomer conversion as continued radical generation will cause some loss of the RAFT end-group.

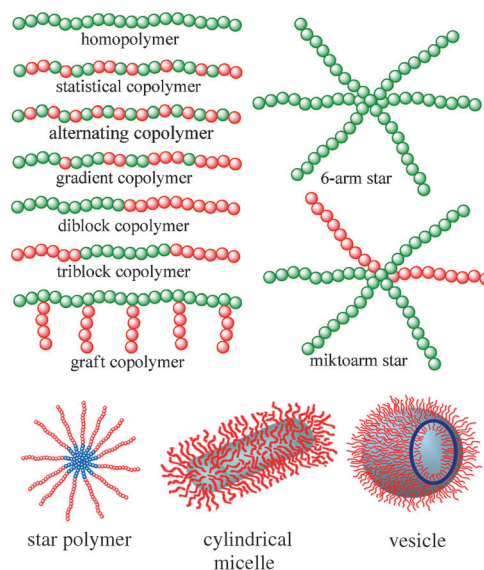


Figure 3. Some polymer structures accessible by RAFT polymerization.

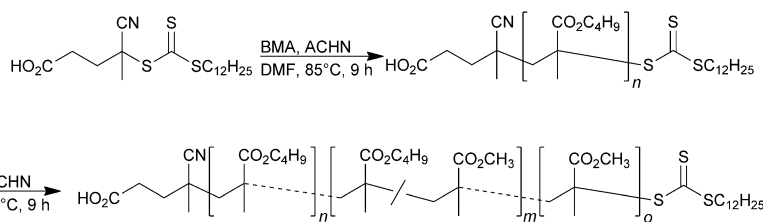
1.2. Continuous Flow RAFT Polymerization

Over the last few years, flow chemistry and microreactor technology have changed the way polymers are synthesized in the laboratory.^[13] The RAFT process can be adapted to allow continuous flow polymer synthesis.^[14] Work in our laboratory has shown that stainless steel tubular reactors are beneficial because of the air sensitivity of the process.^[14a–d] The small dimensions of these reactors, typically 1 mm internal diameter, ensures that the reaction can be conducted

under quasi-isothermal conditions and that the residence time distributions are very narrow with a near plug-flow profile.

In recent work, it has been shown that several unit operations can be combined in series to form an integrated process, namely inline degassing of monomer stock solutions, RAFT

polymerization, RAFT end-group removal, and polymer precipitation and isolation (Figure 4a).^[14b,d] Strategies have also been designed to allow the synthesis of “quasi-block” copolymers (Figure 4b).^[14c] In forming macro-RAFT agents with highest end-group fidelity, polymerization should be taken to less than full (around 90%) monomer conversion. This creates a need to isolate and purify the poly(A) macro-RAFT agent before proceeding with block copolymer synthesis. Alternatively, if the small amount of residual monomer A is not removed, a gradient copolymer will be formed in the chain extension step with monomer B. The length of this segment is dependent on the amount of residual monomer A and the reactivity ratios of A and B in copolymerization. We have termed the products, which have the structure poly(A)-*block*-poly(A-*grad*-B),



Scheme 4. Synthesis of poly(BMA)-*block*-poly(BMA-*grad*-MMA), a “quasi-block” copolymer.

“quasi-block” copolymers. Often the reactivity ratios are such that the residual monomer will be rapidly consumed and the gradient segment will be short. The structure is then poly(A)-*block*-poly(A-*grad*-B)-*block*-poly(B), and the properties are likely to be similar to those of pure poly(A)-*block*-poly(B). An example of quasi-block synthesis is provided in Scheme 4.^[15]

1.3. High-Throughput RAFT Polymerization

Automated parallel synthesizers (APS) such as the Chem-speed or Symyx platforms have been extensively used to perform RDRP,^[16] including nitroxide-mediated polymerization (NMP),^[16b] atom transfer radical polymerization (ATRP),^[16c,d] and RAFT.^[16e] These and other studies of oxygen-sensitive polymerizations in the APS have invariably made use of inert gas sparging techniques to eliminate oxygen from the reaction mixtures. While this method can be effective, it is also one of the more likely sources of irreproducible results.^[17] A further drawback of inert gas sparging is that the concentration of reagents can change when the reagents or solvents are volatile. Thus, an effective automated freeze–evacuate–thaw technique suitable for parallel synthesizers has been developed. The results indicate that the method is as effective as “one at a time” high vacuum freeze–evacuate–thaw degassing using a traditional vacuum line.^[18] We have applied this degassing methodology in the screening and evaluation of switchable RAFT agents^[10c,d] and in the construction of block copolymer libraries used in our studies of siRNA delivery mentioned below.^[19]

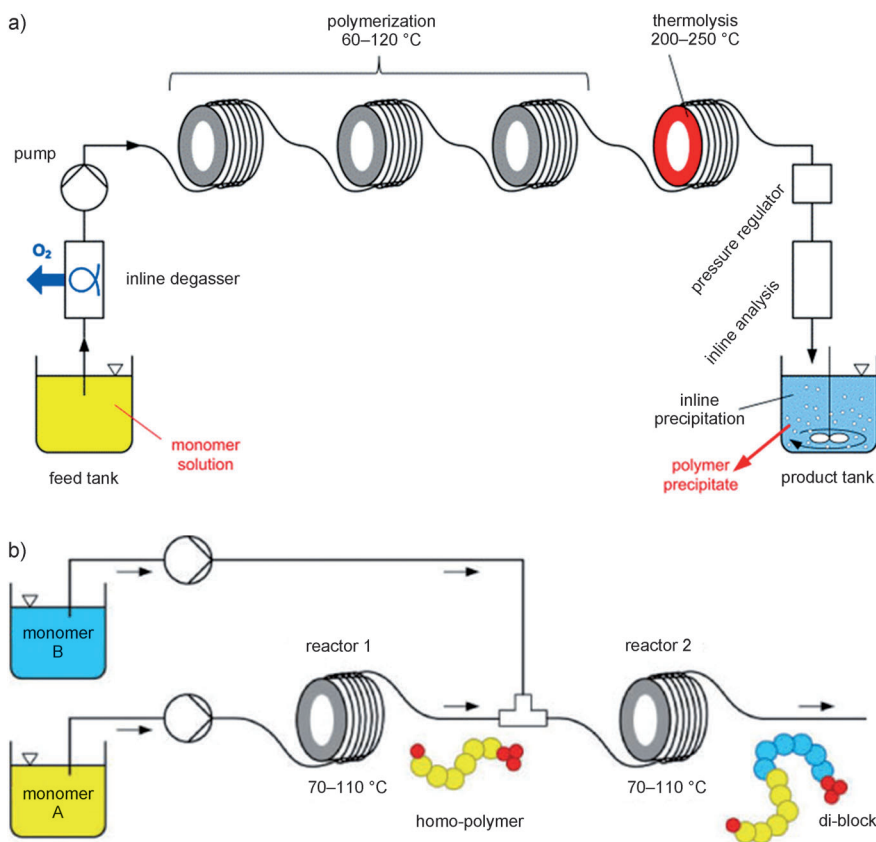


Figure 4. Schematics showing polymer synthesis in a continuous flow process using a sequence of tubular reactors. Adapted from refs. [14b,d].

A convenient (unattended and fully automated) synthesis of libraries of “quasi-block” copolymers by RAFT polymerization with sequential monomer addition has been devised.^[15] The process, illustrated in Figure 5 and Scheme 4,

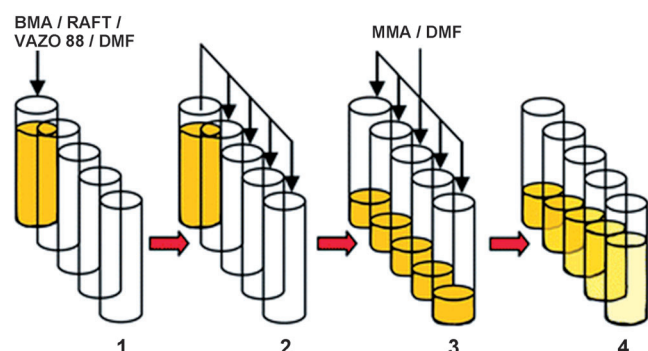


Figure 5. Schematic representation of the automated parallel synthesis of a quasi-diblock copolymer library by sequential monomer addition. Adapted from ref. [15].

involves the initial synthesis of a series of three PBMA macro-RAFT agents (PBMA = poly(*n*-butyl methacrylate)). These macro-RAFT agents were not isolated; rather, the high-conversion polymerization mixtures were distributed equally between a further series of five reactors by the liquid handling system. After the addition of aliquots of MMA, solvent, and initiator, the standard degassing protocol was applied prior to the chain-extension step. The polymerization mixtures were sampled at the various stages of the operation so that extents of the reaction could be established. Characterization by differential scanning calorimetry showed the products, poly(BMA)-*block*-poly(BMA-*grad*-MMA) “quasi-block” copolymers, to have properties very similar to those expected for the corresponding “pure” diblock copolymers of similar overall composition.^[15]

2. RAFT Applications

Since 1998 (the year of filing of the first RAFT patents^[20]), more than 500 patent applications pertaining to RAFT polymerization have been filed by more than 100 companies, and the rate of patenting continues to increase. A survey of granted US patents in the field of RAFT is included in our recent reviews.^[4,21] These patents cover a multitude of applications including polymer therapeutics, drug delivery, biosensors, plastic solar cells, microelectronics, desalination membranes, cosmetics, lubricants, surfactants, paints, inks, and adhesives. In this section we provide a summary of recent research in three specific areas where RAFT polymerization has been applied in developing new materials for optoelectronics, therapeutic delivery, and rheology control.

2.1. Optoelectronic Polymers

Optoelectronic polymers include polymers for hole/electron transport in organic photovoltaics (OPVs), in organic and polymer light-emitting diodes (OLEDs and PLEDs), in thin-film transistors (TFTs), in sensors, in light-harvesting, and related applications. A significant advantage of RAFT polymerization over many alternate processes in the context of optoelectronic applications is that no undesired metal species are introduced during the polymerization process.^[22]

The fully conjugated polymers that have seen use in organic semiconductors cannot be directly made by the RAFT process or other RDRP methods. Nonetheless, RDRP can be used in the synthesis of materials that form one or more of the active components of optoelectronic devices such as photovoltaic cells (Figure 6). Specifically, these methods can

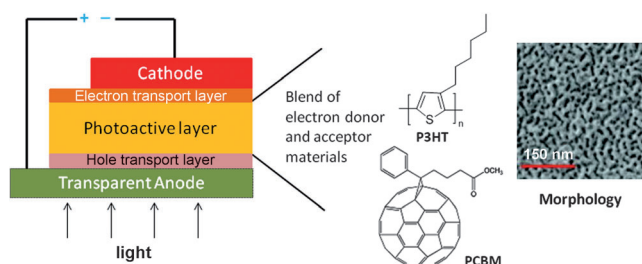


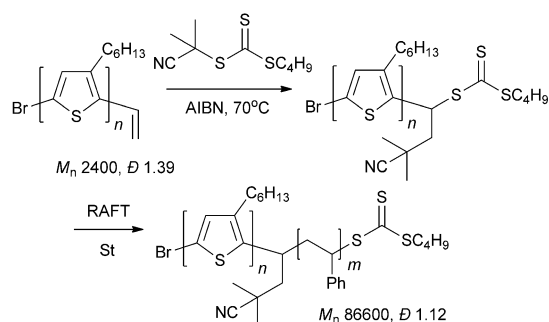
Figure 6. Schematic representation of a photovoltaic cell in which poly(3-hexylthiophene) (P3HT) and [6,6]-phenyl-C₆₁-butyric acid methyl ester (PCBM) are incorporated in the photoactive layer. The scanning electron micrograph (SEM) image shows the morphology of the photoactive layer.

be used to form polymers that contain segments of these polymers either as blocks^[22,23] or pendant units.^[22,24]

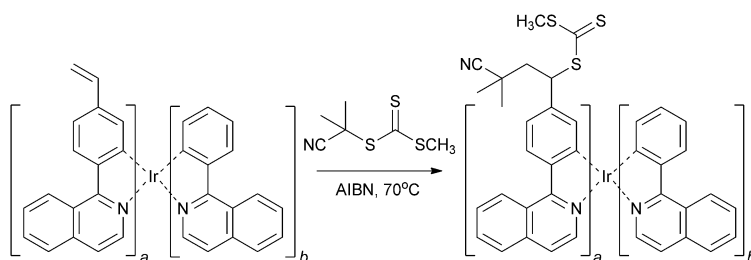
Block copolymers have attracted interest because of their ability to self-assemble to give nanophase separation into periodic domains. The dimensions of these domains can be in the range of 5–50 nm, which encompasses that required for many semiconductor applications. Block copolymers may also be added as a minor component to control the morphology of a blend by acting as a compatibilizer or structure director.

In using RAFT to design block copolymers, it is important that block linkage is 1) short and 2) does not contain potentially hydrolyzable groups (such as ester or amide units), and that 3) the RAFT functionality is not part of the linkage.^[23b,25] Transformation of a vinyl compound to a macro-RAFT agent by RAFT single unit monomer insertion^[26] provides a route to such ‘R’-connected block copolymers. The method is applicable to MAMs such as vinylthiophene and was demonstrated with the preparation of block copolymers containing organic semiconductor segments. Thus, a poly(3-hexylthiophene) block copolymer was prepared by RAFT single unit monomer insertion as shown in Scheme 5.^[27]

The same procedure was used to prepare a series of macro-RAFT agents from iridium complexes as shown in Scheme 6.^[28] These were used as the dopant core in red



Scheme 5. Macro-RAFT agent synthesis from poly(3-hexylthiophene) macromonomer by single unit monomer insertion and its use in RAFT polymerization to form poly(3-hexylthiophene)-*block*-polystyrene (M_n in polystyrene equivalents).^[27]



Scheme 6. Macro-RAFT agent synthesis from iridium complex by single unit monomer insertion ($a = 1,2,3$, $b = 2,1,0$).^[28]

phosphorescent, solution-processable, and chain-end functional star polymers that had arms based on a functionalized styrene as host. The star metallopolymers were in turn a key component of PLEDs.

Another approach was applied in the synthesis of a donor–acceptor block copolymer which comprised a fully-conjugated rigid-rod polymer (F6T2) and a coil polymer with pendant C₆₀ functionality.^[29] In the first step, the synthesis of polystyrene with a terminal bromophenyl and pendant-protected amine functionality was synthesized by RAFT polymerization. The F6T2 segment with a terminal borinate ester functionality was then prepared. These fragments were then subjected to Suzuki coupling to form a tri-block copolymer, and finally the [6,6]-phenyl-C₆₁-butyric acid methyl ester (PCBM) groups were attached (Scheme 7).

The morphology of the block copolymer exhibited a substantially improved stability compared to a blend with a similar overall composition, thus suggesting the material to be potentially useful in improving the performance of organic photovoltaics.^[29] The blend at room temperature exhibited a higher surface roughness that decreased upon annealing at 150°C (2 × 10 min). On the other hand, the block copolymer displayed a smoother surface at room temperature that remained essentially unchanged after annealing under similar conditions.

Grafting to poly(chloromethylstyrene) has also been a feature in other syntheses, although the methods of grafting

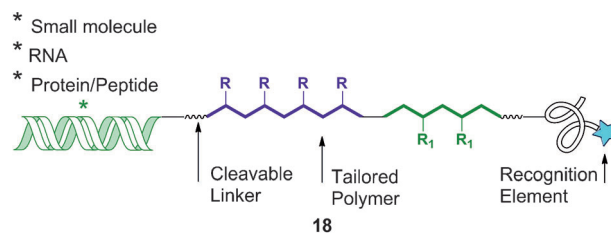
vary.^[22,30] Conjugation after RAFT polymerization by using a Wittig reaction has been done to prepare a polystyrene with pendant poly[2-methoxy-5-(2'-ethylhexyloxy)-1,4-phenylenevinylene] (MEH-PPV) chains (Scheme 8).^[30b] The polymers exhibited tuneable absorption and emission that were dependent on the conjugation length of the pendant MEH-PPV chains and can be considered as a template for producing polymers with controlled absorption and emission for light harvesting and optoelectronic applications.

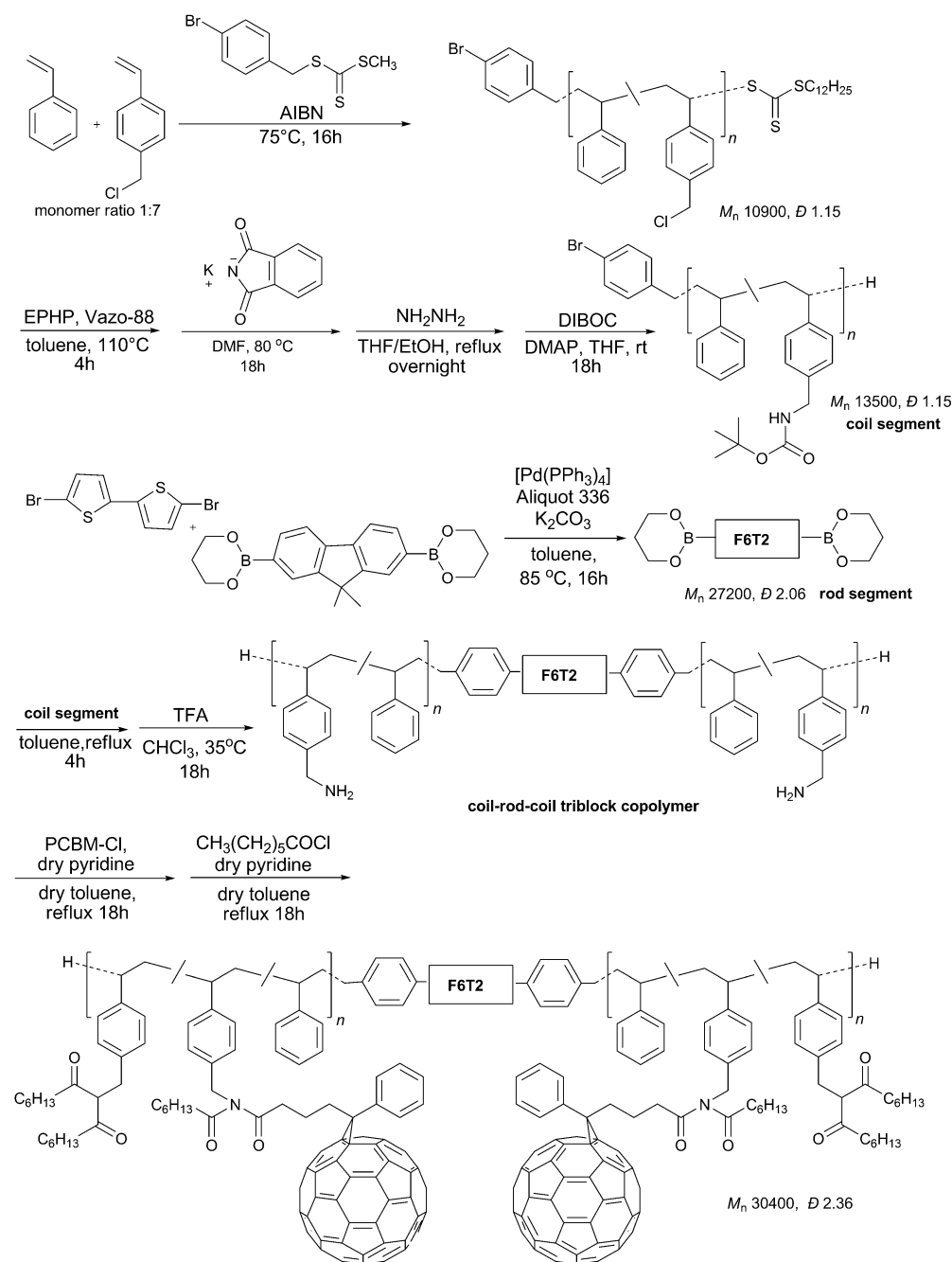
2.2. Therapeutic Delivery

There is substantial interest in the synthesis of block and graft copolymers (conjugates) with drug molecules or biological polymers.^[31] Such polymers can often be obtained through RAFT polymerization mediated by an appropriately designed macro-RAFT agent based on the substrate of interest.^[8,31f] However, the most common methods for forming these materials involve coupling the RAFT-synthesized polymer with the biopolymer through reactive functionality present on either the R or Z groups after RAFT polymerization. Processes used include the active ester–amine reaction and

the thiazolidine-2-thione–amine reaction.^[4] It is important to note that use of functionality on Z leaves the thiocarbonylthio group as a potentially degradable link at the block juncture, which may or may not be advantageous depending on the intended application. Conjugation following RAFT polymerization may also involve direct transformation of the RAFT end-group, often by way of a “thiol-click” reaction such as the pyridyl disulfide–thiol, the thiol–ene, or the thio-Michael reaction.^[31h,32]

We have used RAFT in the synthesis of structures **18** that form the basis of a new platform technology with the potential to deliver bioactives including small organic actives and biomacromolecules (i.e., RNA, DNA, proteins, and peptides). Applications lie in the drug delivery, biomedical, agrochemical, and personal care fields. The advantage of our technology is that it can be readily tailored to meet specific challenges such as altered solubility, controlled delivery, and targeted delivery and release.





Scheme 7. Synthesis of block copolymer comprising conjugated rigid-rod polymer (F6T2) and styrenic coil polymer with pendant C_{60} functionality (M_n in polystyrene equivalents).^[29]

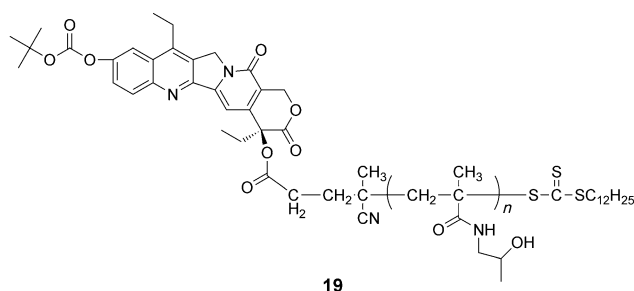
A series of well-defined polymer–drug conjugates **19** were prepared to modify the physical properties of a known cytotoxic anticancer drug,^[33] SN-38. RAFT polymerization was used to site-specifically append a *N*-(2-hydroxypropyl) methacrylamide (HPMA) in a grafting-from process. The poly(HPMA)–SN-38 conjugates displayed excellent aqueous solubility and stability, whilst retaining the cytotoxic activity of the parent drug. In vitro co-culture assays with both cancer and non-cancer cell lines demonstrated specificity

of RAFT-derived poly-(HPMA)–SN-38 conjugates for cancerous cells.

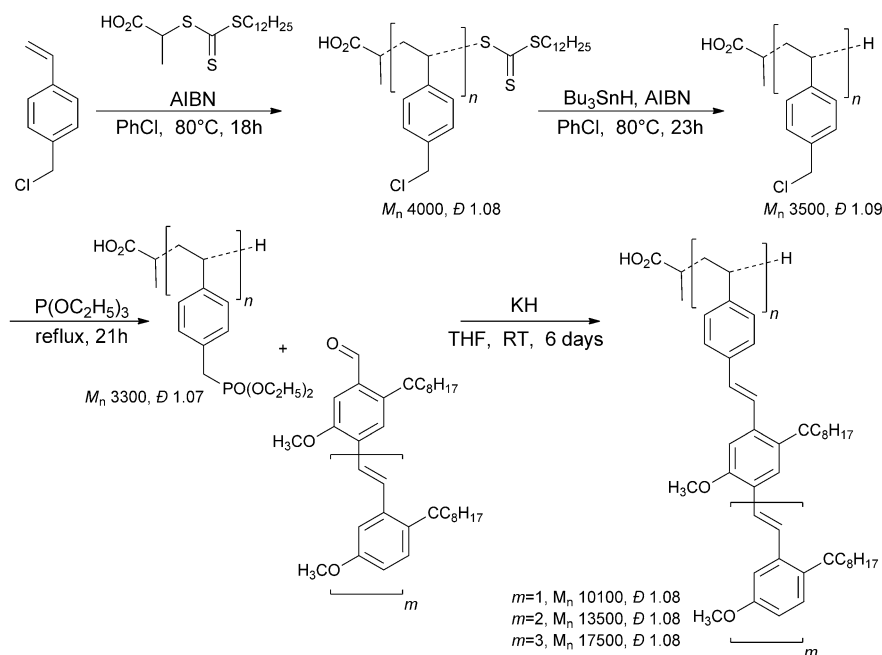
A series of ABA tri-block copolymers **20** were prepared from oligo(ethylene glycol) methyl ether methacrylate (OEGMA475) and *N,N*-dimethylaminoethyl methacrylate (DMAEMA) to investigate the effect of polymer composition on cell viability, siRNA uptake, serum stability, and gene silencing.^[19]

Eight block copolymers were prepared by systematically varying the central cationic block (DMAEMA) length from 38 to 192 monomer units and the outer hydrophilic block (OEGMA475) from 7 to 69 units. The results indicate that the length of the central cationic block is the key structural parameter determining cell viability and gene silencing efficiency. Block lengths of 110 to 120 monomer units proved optimal. The ABA block copolymer architecture is also critical. The outer hydrophilic blocks contribute to serum stability and overall efficiency of the polymer as a delivery system (Figure 7).

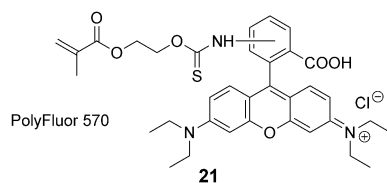
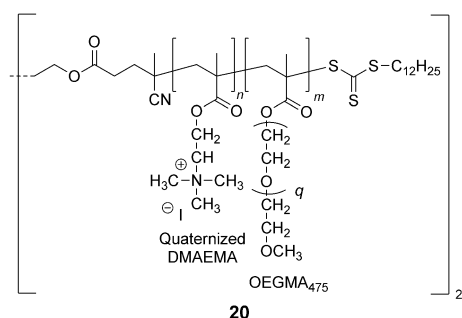
A similar ABA triblock was prepared with Poly-Fluor570 (**21**) as a minor component in the OEGMA475 block in the second step of the polymer synthesis. This allowed us to study cellular uptake and to intracellularly track the



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Scheme 8. Synthesis of a graft copolymer comprising a styrenic backbone and pendant (MEH-PPV) chains ($m=1-3$, M_n in polystyrene equivalents).^[30b]



polymer-siRNA complex in CHO-GFP cells and Huh7-GFP cells (Figure 8).^[19b]

We have extended our siRNA delivery study with star-copolymer architectures made by the RAFT process.^[34] A unique feature of these star copolymers is that the core is dual-degradable due to the presence of disulfide and ester linkages. The structure of the 4-arm RAFT agent and the star copolymer-siRNA complex are shown in Scheme 9.

2.3. Rheology Control Agents

Two of the more popular methods for forming core-cross-linked star polymers involve RAFT (co)polymerization of a di- (or higher) functional monomer to form a core.^[35] The

so-called 'arm-first' process for star synthesis makes use of a linear macro-RAFT agent (the arm) to mediate microgel formation. Another approach relies on the RAFT agent functionality retained in a RAFT-synthesized microgel to enable arms to be grown from the core in a subsequent RAFT polymerization of a mono-ene. The two approaches may be combined to synthesize miktoarm stars. The polymers have found a range of applications in the fields of drug delivery, dye/pigment encapsulation, and rheology control.

Researchers at Lubrizol^[36] have applied the arm-first approach (Scheme 10) in the development of their Asteric methacrylate-based viscosity modifiers for lubricating fluids. The star polymers exhibit an

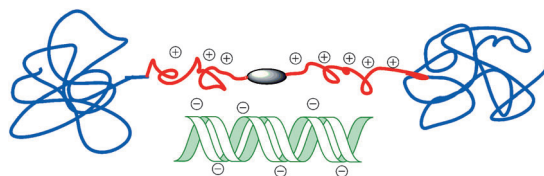


Figure 7. Schematic representation of the interaction between the positively charged midblock of copolymers **20** and negatively charged siRNA.

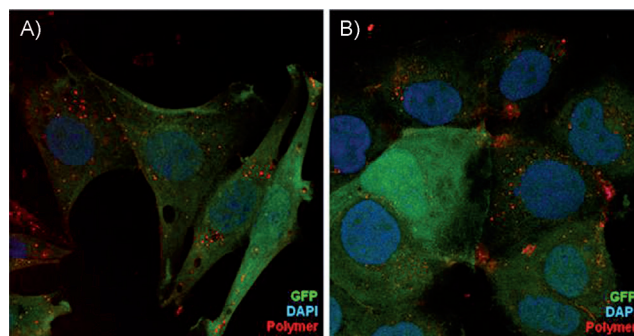
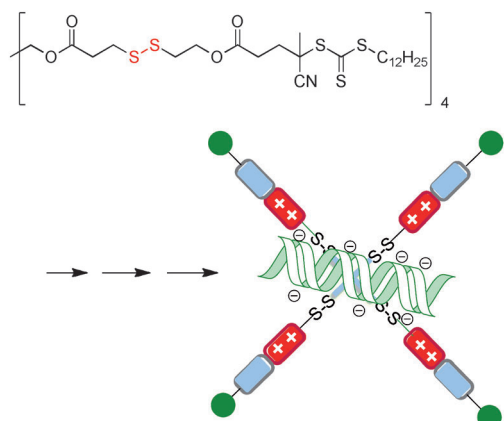
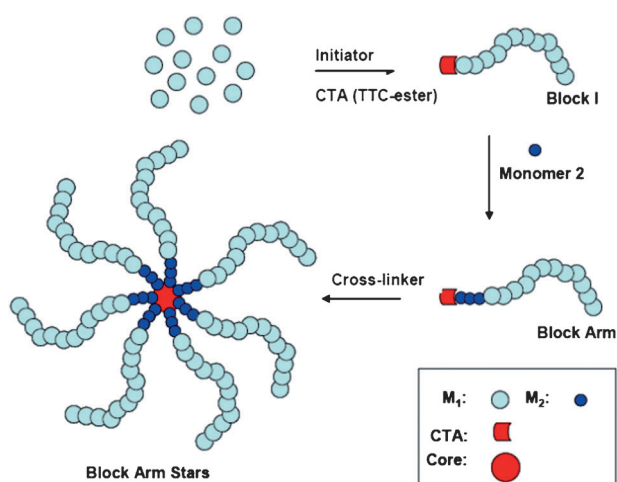


Figure 8. Confocal scanning microscopy images demonstrating the cellular uptake of a red fluorescent RAFT polymer-siRNA complex in which PolyFluor 570 (**21**) had been incorporated. A) CHO-GFP cells (CHO, Chinese hamster ovary cell line); B) Huh7-GFP cells (Huh7, a hepatocarcinoma cell line). GFP, green fluorescent protein. Cell nuclei were stained with DAPI (4',6-diamidino-2-phenylindole, a blue fluorescent stain that binds strongly to DNA). The red dots are fluorescent signals from the RAFT polymer-siRNA complex. After endocytosis, they can be seen within the endosome compartments. Reproduced from ref. [19] with permission.



Scheme 9. Tetra-RAFT agent used to make a 4-armed star for siRNA delivery.



Scheme 10. Arm-first approach to star-shaped molecules. Reproduced from [36] with permission.

improved viscosity index (less viscosity change with temperature) and shear stability at comparable thickening versus linear polymers. High viscosity index fluids are well known to improve vehicle fuel economy.

Conclusions and Perspective

With appropriate selection of the RAFT agent for the monomers and reaction conditions, RAFT polymerization is a robust and versatile process applicable to the majority of monomers subject to radical polymerization. While it has not been a topic of this review, there remains significant research focus on the development of yet more versatile RAFT agents to provide even greater flexibility in the construction of novel architectures,^[8] on improving the precision to control such features as monomer sequence distribution^[26b] and polymer stereochemistry,^[37] and on devising efficient processes for RAFT end-group transformation following RAFT polymerization.^[32,38]

However, most efforts in the field of RAFT polymerization (and other RDRP methods) at CSIRO and elsewhere are now clearly directed towards applications. The three areas discussed herein are of much current and ongoing interest. Other applications areas that are the subject of current research have been mentioned above.

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